

WEEK 3 PROJECT REPORT

FINANCIAL DATA IMPORT, TRANSFORMATION & TIME INTELLIGENCE

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Project Title: Shopify Stock Market Analysis using Power BI

Tool Used: Microsoft Power BI Desktop

Dataset: Shopify Stock Data

Introduction

This project focuses on analyzing Shopify stock market data using **Microsoft Power BI**. The objective is to import, transform, model, and visualize historical stock data to understand changes in stock prices and trading activity over time.

The project uses Power BI features such as **Power Query, DAX, Date Tables, relationships, calculated measures, time intelligence, charts, KPI cards and slicers** to create an interactive financial dashboard.

Objectives

The main objectives of this project are:

- To import Shopify stock data into Power BI.
- To prepare and transform the stock data using Power Query.
- To create a dedicated Date Table using DAX.
- To establish a relationship between the stock dataset and Date Table.
- To create important financial measures using DAX.
- To calculate daily stock returns.
- To analyze stock prices over different time periods.
- To apply time-intelligence functions.
- To create Year, Quarter, Month and Day hierarchy.
- To build an interactive dashboard using charts, KPI cards and slicers.

Dataset

The dataset used in this project is **Shopify Stock Data**.

The dataset contains historical stock-market information, including:

- Date
- Open Price

- High Price
- Low Price
- Close Price
- Adjusted Close Price
- Trading Volume

This data allows the stock performance of Shopify to be analyzed across different dates and time periods.

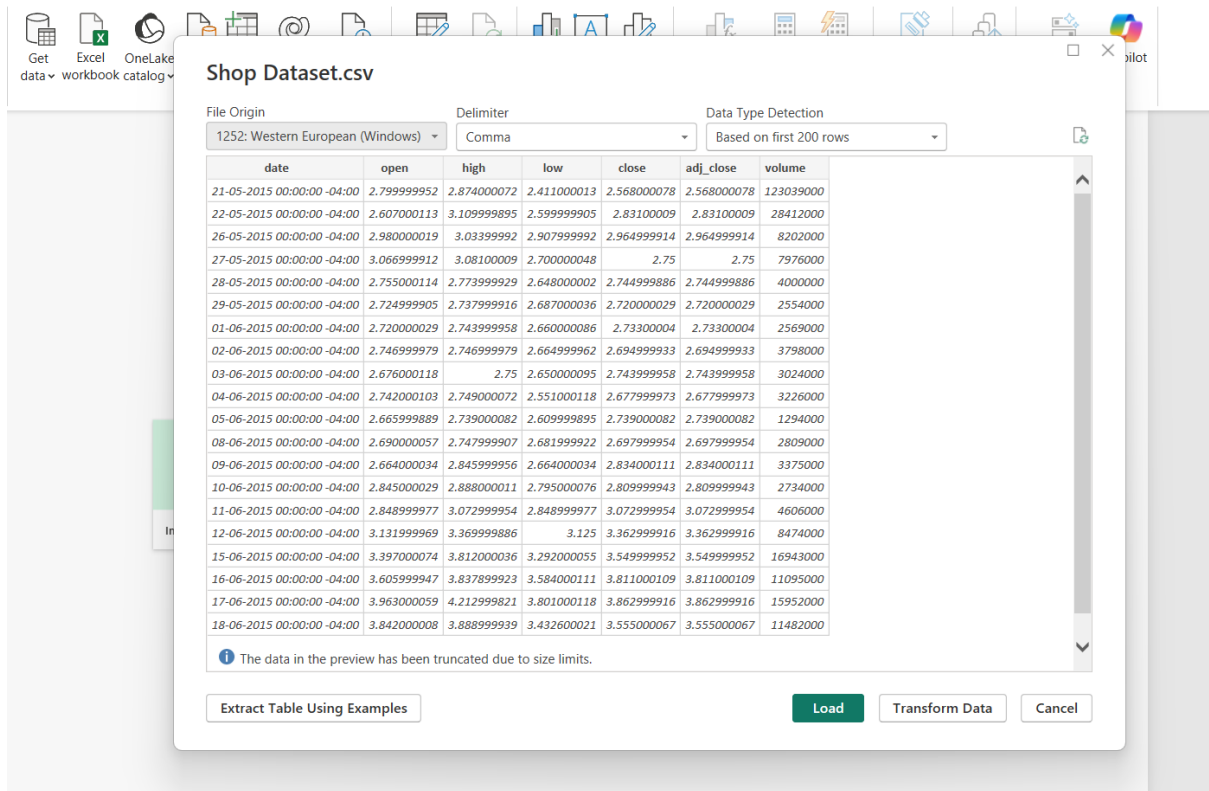
Table.TransformColumnTypes("#Promoted Headers",({date", type datetimezone), {"open", type number), {"high", type number), {"low", type number), {"close", type number), {"adj_close", type number), {"volume", type number})										
	date	1.2 open	1.2 High	1.2 low	1.2 close	1.2 adj_close	1.2 volume			
		Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	21-05-2015 00:00:00 -04:00	2.799999952	2.874000072	2.411000013	2.568000078	2.568000078	2.831000009			
2	22-05-2015 00:00:00 -04:00	2.607000113	3.109999895	2.599999905	2.831000009	2.831000009	2.733000004			
3	26-05-2015 00:00:00 -04:00	2.980000019	3.033999992	2.907999992	2.964999914	2.964999914	2.75			
4	27-05-2015 00:00:00 -04:00	3.066999912	3.081000009	2.700000048	2.75	2.75	2.744999886			
5	28-05-2015 00:00:00 -04:00	2.755000114	2.773999929	2.648000002	2.744999886	2.744999886	2.720000029			
6	29-05-2015 00:00:00 -04:00	2.724999905	2.737999916	2.687000036	2.720000029	2.720000029	2.733000004			
7	01-06-2015 00:00:00 -04:00	2.720000029	2.743999958	2.660000086	2.733000004	2.733000004	2.694999933			
8	02-06-2015 00:00:00 -04:00	2.746999979	2.746999979	2.664999962	2.694999933	2.694999933	2.743999958			
9	03-06-2015 00:00:00 -04:00	2.676000118	2.75	2.650000095	2.743999958	2.743999958	2.677999973			
10	04-06-2015 00:00:00 -04:00	2.742000103	2.749000072	2.551000118	2.677999973	2.677999973	2.739000082			
11	05-06-2015 00:00:00 -04:00	2.665999889	2.739000082	2.609999895	2.739000082	2.739000082	2.697999954			
12	08-06-2015 00:00:00 -04:00	2.690000057	2.747999907	2.681999922	2.697999954	2.697999954	2.834000111			
13	09-06-2015 00:00:00 -04:00	2.664000034	2.845999956	2.664000034	2.834000111	2.834000111	2.809999943			
14	10-06-2015 00:00:00 -04:00	2.845000029	2.888000011	2.795000076	2.809999943	2.809999943	3.125			
15	11-06-2015 00:00:00 -04:00	2.848999977	3.072999954	2.848999977	3.072999954	3.072999954	3.362999916			
16	12-06-2015 00:00:00 -04:00	3.131999969	3.369999886	3.125	3.362999916	3.362999916	3.549999952			
17	15-06-2015 00:00:00 -04:00	3.397000074	3.812000036	3.292000055	3.549999952	3.549999952	3.811000109			
18	16-06-2015 00:00:00 -04:00	3.605999947	3.837899923	3.584000111	3.811000109	3.811000109	3.862999916			
19	17-06-2015 00:00:00 -04:00	3.963000059	4.212999821	3.801000118	3.862999916	3.862999916	3.555000067			
20	18-06-2015 00:00:00 -04:00	3.842000008	3.888999939	3.432600021	3.555000067	3.555000067	3.552000046			
21	19-06-2015 00:00:00 -04:00	3.632999897	3.674000025	3.255000114	3.552000046	3.552000046	3.694000006			
22	22-06-2015 00:00:00 -04:00	3.676000118	3.849999905	3.608999968	3.694000006	3.694000006	3.542999983			
23	23-06-2015 00:00:00 -04:00	3.775000095	3.809999943	3.516999996	3.542999983	3.542999983	3.595999956			
24	24-06-2015 00:00:00 -04:00	3.512000084	3.742899895	3.414000034	3.595999956	3.595999956	3.483999968			
25	25-06-2015 00:00:00 -04:00	3.634000063	3.650000095	3.480000019	3.483999968	3.483999968				
26										

Data Import and Preparation

The Shopify stock dataset was imported into **Power BI**.

The data was inspected and prepared before creating the dashboard. The date column was used as the primary time-related field for analyzing stock performance.

The dataset was connected to Power BI's data model so that calculations and visualizations could be created using DAX.



Date Dimension Table

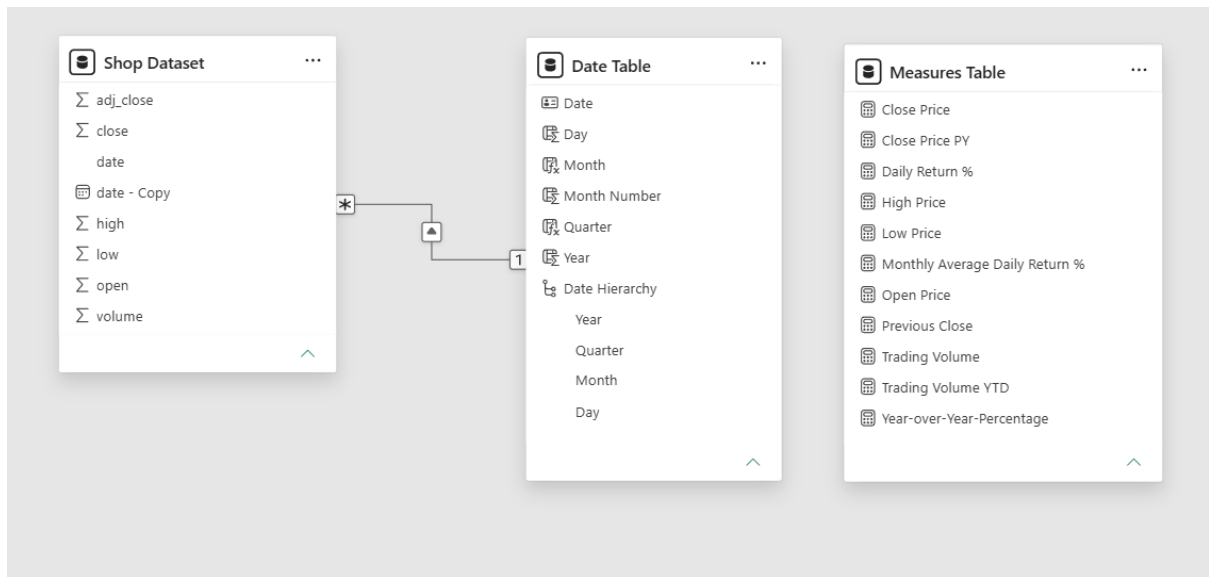
A dedicated **Date Table** was created to support time-based analysis.

The Date Table contains the following fields:

- Date
- Day
- Month
- Month Number
- Quarter
- Year

The Date Table provides a structured calendar that can be used for time-intelligence calculations.

The **Month Number** field was also used to ensure that months are displayed in the correct chronological order rather than alphabetical order.



Date Table and Relationship

The Date Table was connected to the **Shop Dataset** using the Date field.

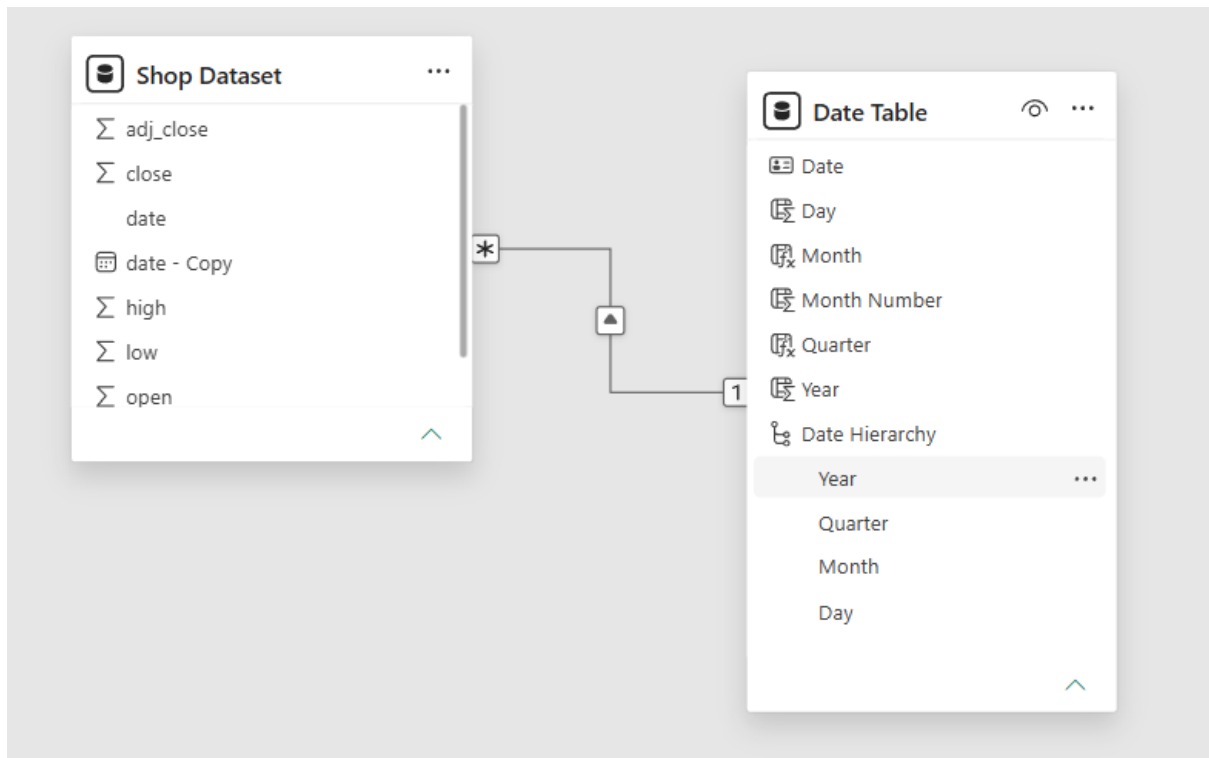
The relationship created was:

Shop Dataset[date] → Date Table[Date]

The relationship is:

- **Many-to-One (*:1)**
- **Active**
- **Single-direction filtering**

This relationship allows the Date Table to filter the stock data correctly and supports time-intelligence calculations.



Date Hierarchy

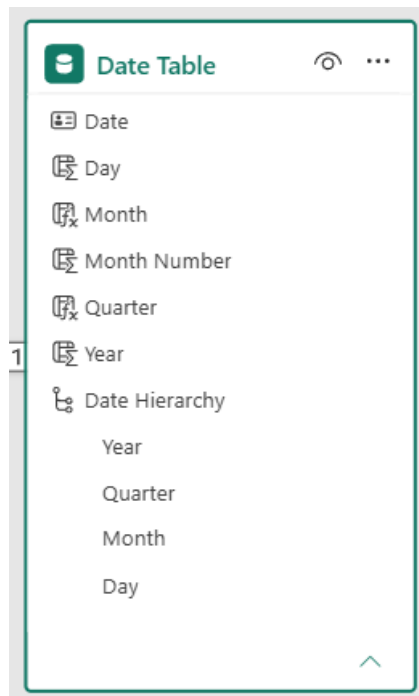
A custom **Date Hierarchy** was created for time-based analysis.

The hierarchy contains:

1. Year
2. Quarter
3. Month
4. Day

This hierarchy makes it possible to analyze stock performance at different levels of time.

For example, the dashboard can show stock prices by year and can also drill down into quarter, month, and day.



DAX Measures

Several DAX measures were created to analyze Shopify's stock performance.

1. Open Price

Open Price = MAX('Shop Dataset'[open])

This measure displays the opening stock price.

2. Close Price

Close Price = MAX('Shop Dataset'[close])

This measure displays the closing stock price.

3. High Price

High Price = MAX('Shop Dataset'[high])

This measure displays the highest stock price.

4. Low Price

Low Price = MIN('Shop Dataset'[low])

This measure displays the lowest stock price.

5. Trading Volume

Trading Volume = SUM('Shop Dataset'[volume])

This measure calculates the total trading volume.

6. Previous Close

A **Previous Close** measure was created to obtain the closing price from the previous available trading date.

Previous Close =

VAR CurrentDate =

MAX('Date Table'[Date])

VAR PreviousTradingDate =

```
CALCULATE(
    MAX('Shop Dataset'[date]),
    FILTER(
        ALL('Shop Dataset'[date]),
        'Shop Dataset'[date] < CurrentDate
    )
)
```

RETURN

```
CALCULATE(
    MAX('Shop Dataset'[close]),
    REMOVEFILTERS('Date Table'),
    'Shop Dataset'[date] = PreviousTradingDate
)
```

This calculation uses the previous available trading date instead of simply using the previous calendar day.

This is important because stock markets do not normally have trading data for weekends and market holidays.

Daily Return %

The Daily Return % measure was created to calculate the percentage change between the current closing price and the previous trading day's closing price.

Daily Return % =

VAR CurrentClose =

```
MAX('Shop Dataset'[close])
```

```
VAR PreviousCloseValue =
```

```
[Previous Close]
```

```
RETURN
```

```
IF(
```

```
ISBLANK(CurrentClose) ||
```

```
ISBLANK(PreviousCloseValue),
```

```
BLANK(),
```

```
DIVIDE(
```

```
CurrentClose - PreviousCloseValue,
```

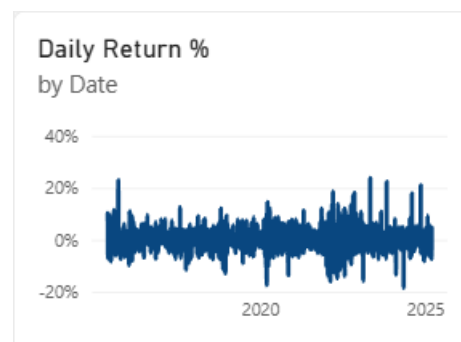
```
PreviousCloseValue
```

```
)
```

```
)
```

The result shows the daily percentage movement of Shopify's stock.

Positive values represent an increase, while negative values represent a decrease.



Monthly Average Daily Return %

A monthly average return measure was also created to summarize daily returns by month.

Monthly Average Daily Return % =

```
AVERAGEX(
```

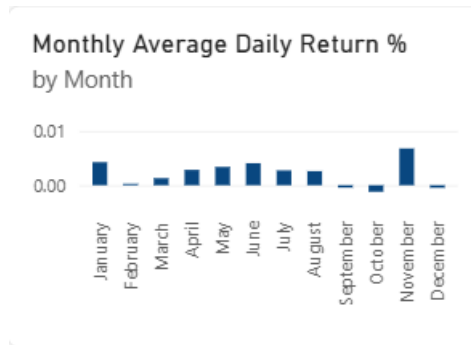
```
VALUES('Date Table'[Date]),
```

```
[Daily Return %]
```

```
)
```

This measure was visualized using a column chart with **Month** on the X-axis.

The chart provides a monthly view of average daily stock returns.



Time Intelligence

Time-intelligence calculations were included in the project to analyze stock performance across different time periods.

The project includes measures related to:

- Previous Year Close Price
- Year-to-Date Trading Volume
- Year-over-Year Percentage

These calculations help compare financial performance across different periods.

Close Price Analysis

A line chart was created using **Close Price** and **Year**.

The chart shows how Shopify's closing stock price changed over the historical period.

The visualization makes it easier to identify major increases and decreases in the stock price.



Open Price and Close Price Analysis

A line chart was created to compare:

- Open Price
- Close Price

The chart helps visualize the relationship between opening and closing prices across different years.



High Price and Low Price Analysis

A line chart was created using:

- High Price
- Low Price

This visualization shows the range between the highest and lowest stock prices over time.

It helps demonstrate periods where Shopify experienced larger changes in its stock price.

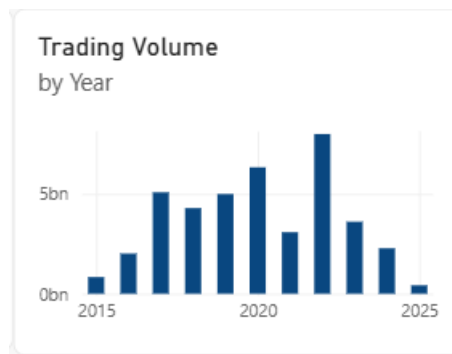


Trading Volume Analysis

A column chart was created to display **Trading Volume by Year**.

Trading volume represents the amount of stock traded during the selected period.

The chart allows the trading activity of Shopify to be compared across different years.



Year-over-Year Analysis

A Year-over-Year comparison was included to compare the current period with the corresponding previous-year period.

The dashboard includes a comparison involving:

- Close Price
- Previous Year Close Price

This makes it possible to examine how Shopify's stock price changed compared with the previous year.

[Insert Screenshot: Year-over-Year Comparison]

Dashboard KPI Cards

The dashboard contains KPI cards for the main stock indicators.

The cards display:

- **Open Price:** 48.46
- **High Price:** 176.29
- **Low Price:** 1.85
- **Close Price:** 48.46
- **Trading Volume:** 41bn

These cards provide a quick summary of the selected stock data.

Open Price 48.46	High Price 176.29	Low Price 1.85	Close Price 48.46	Trading Volume 41bn
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Interactive Slicers

Interactive slicers were added to the dashboard to allow users to filter the data.

The dashboard contains slicers for:

Year

Users can select a specific year or multiple years.

Quarter

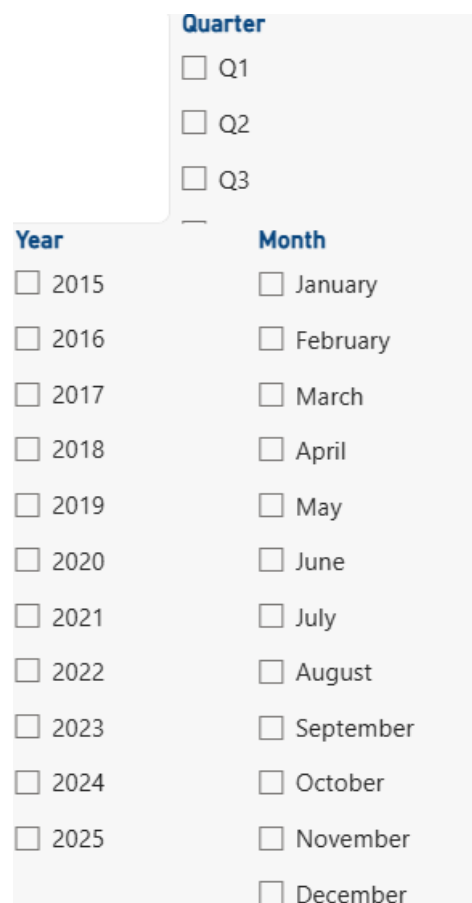
Users can select:

- Q1
- Q2
- Q3
- Q4

Month

Users can select individual months such as January, February, March, and so on.

These slicers make the dashboard interactive and allow users to explore specific periods.



The image shows a screenshot of three interactive slicers. The 'Quarter' slicer is at the top, with a title 'Quarter' in blue. It contains three options: 'Q1', 'Q2', and 'Q3', each with an unchecked checkbox. Below it is the 'Year' slicer, with a title 'Year' in blue, and a list of years from 2015 to 2025, each with an unchecked checkbox. To the right of the 'Year' slicer is the 'Month' slicer, with a title 'Month' in blue, and a list of months from January to December, each with an unchecked checkbox. The slicers are presented in a light gray, semi-transparent overlay.

Quarter
☐ Q1
☐ Q2
☐ Q3

Year
☐ 2015
☐ 2016
☐ 2017
☐ 2018
☐ 2019
☐ 2020
☐ 2021
☐ 2022
☐ 2023
☐ 2024
☐ 2025

Month
☐ January
☐ February
☐ March
☐ April
☐ May
☐ June
☐ July
☐ August
☐ September
☐ October
☐ November
☐ December

Dashboard Design

The final dashboard was designed with a clean and professional layout.

The dashboard includes:

- A title section
- Subtitle
- KPI cards
- Stock price charts
- Trading volume chart
- Daily return chart
- Monthly average return chart
- Year slicer
- Quarter slicer
- Month slicer

A consistent blue theme was used throughout the dashboard to provide a professional financial-analysis appearance.



Conclusion

The **Shopify Stock Market Analysis** project demonstrates how Power BI can be used to transform historical stock-market data into an interactive financial dashboard.

The project covered the complete process of **data import, data preparation, data modeling, Date Table creation, relationship building, DAX calculations, time intelligence, and dashboard development**.

The final dashboard provides a visual representation of Shopify's stock prices, trading volume, daily returns, and monthly return patterns. The use of KPI cards, charts, and interactive slicers makes the dashboard useful for exploring stock performance across different time periods.

Overall, this project demonstrates the practical application of **Power BI and DAX for financial data analysis and time-series visualization**.